



BESS Dismantling Technical Description (Gen8-10MWh)

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Doc No.: PMD-ES000009	Rev.: B	State: released	Classification: Internal

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Disclaimer

All technical data is subject to change in line with ongoing technical development. All the responsibilities and obligations are subject to the terms of the contract.

The illustrated figures in this documentation do not necessarily reflect the exact scope of supply. The actual scope of supply is subject to technical alterations at any time.

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1 Introduction

This document describes how to dismantle the Battery Energy Storage System (BESS) on-site, including mechanical and electrical dismantling of the DC and AC systems.

2 Safety



Safety is the prerequisite for all the site work.

To ensure the safety of operators and BESS equipment, all personnel involved in site work must carefully read and strictly abide by safety specifications. It is the worker's responsibility to meet all safety, local, and general regulations that apply for the specific site, task, or project.

For the specific safety requirements of different processes, refer to the related delivery documentation during contract performance phase.

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3 Special Tool List

No.	Tool Description	Specification	Quantity	Unit
1	Lifting beam for energy storage system		1	Set
2	Round sling	40 t-3 m	2	Set
3	Shackle	55 t	4	Set
4	Round sling	20t-15m	2	Set
5	Round sling	15t-8m	2	Set
6	Shackle	17 t	4	Set
7	Container hook	CH-15,15 t	1	Set
8	Round sling	15t-9m	1	Set
9	Round sling	40t-3m	2	Set
10	Round sling	25t-9m	4	Set
11	Shackle	17t	4	Set
12	Container hook	CH-20,80T	1	Set
13	Torque wrench	14×18, 60—320 N·m, p=±2%	1	Set
14	Gland tightening tooling	Gland tightening tooling	1	Set
15	Torque wrench	NovaTork, torque wrench, 1/2, 0-100NM, +3%, DSM3100	1	Set
16	Adjustable wrench	Adjustable wrench, L = 12 in, Smax = 35	1	Set
17	Megohmmeter	FLUKE,, 1550C 5 kV	1	Set
18	Hand-held digital multimeter	SZBJ, 2000 V voltage measuring multimeter, VC92, VC92.0-2000 V	1	Set



The above list of tools is for reference only. The actual dismantling process might vary with different equipment models.

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4 Product Introduction

4.1 DC System

The DC system is a container-type centralized liquid-cooling system with high strength, strong load-bearing capacity, safety and reliability.

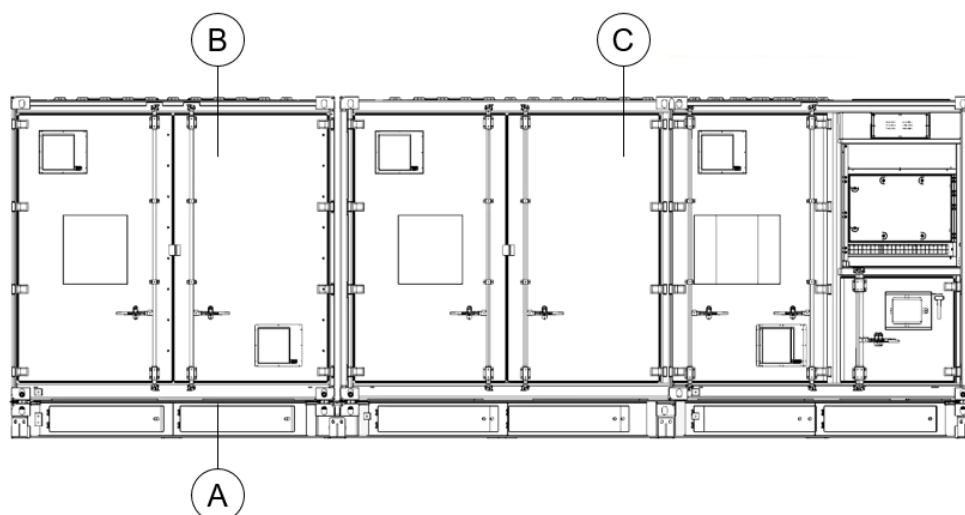


Figure 4-1 DC system

序号	名称	尺寸 (长 x 宽 x 高 mm)	重量 (t)
A	SKID	9199-2438-360	3
B	2 号集装箱	3091-2438-2896	27.5
C	1 号集装箱	6058-2438-2896	44.5



The figure applied here is for reference only. The actual product might vary based on different contracts.

4.2 AC System

The AC system includes equipment such as the Power Conversion System (PCS), transformer, auxiliary transformer, and switch cabinet.

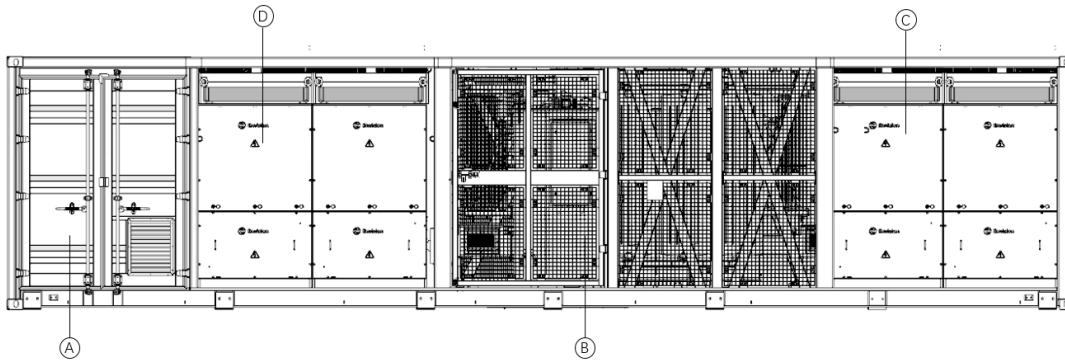


Figure 4-2 AC system

No.	Name	Dimensions (Length x Width x Height, mm)	Weight (t)
A	Switch cabinet	12192 x 2438 x 2896	48.63
B	Transformer		
C	PCS		
D	PCS		



The figure applied here is for reference only. The actual product might vary based on different contracts.

5 Power Off Systems

- 1 Power off the AC system.
- 2 Power off the DC system.



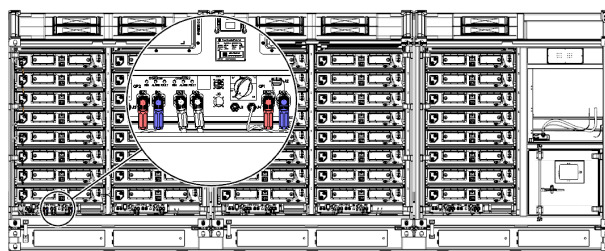
Personnel must wear Level 2 arc flash protection gear when working on the main distribution cabinet and high-voltage box.

- 3 Disconnect the power and perform the Lock Out Tag Out (LOTO) procedure.

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- 4 Disconnect the total negative power cable connectors from all RACKs.

Disconnect the total positive incoming power cable connectors from the Battery Junction Box (RBJB) in the DC container.

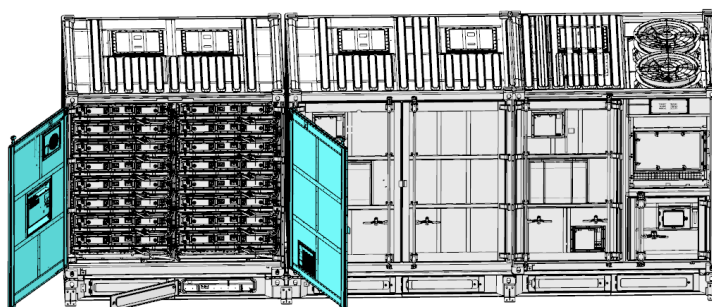


After power disconnection, always verify the voltage before operating any equipment, to avoid electric risk.

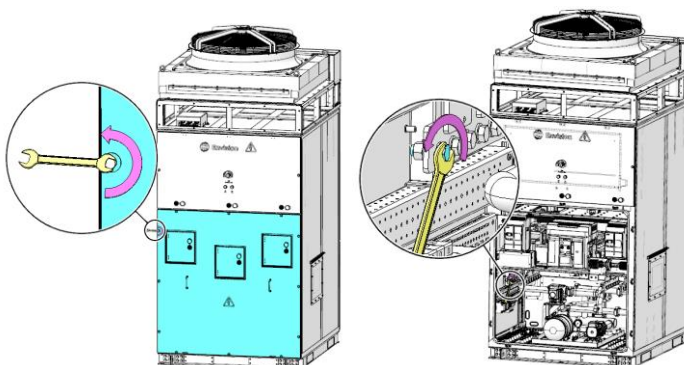
6 Remove equipment cables

6.1 Preparation Before Disconnection

- 1 Open the DC system container door and the cabinet door.



- 2 Remove the bolts from the PCS cover plate of the AC system, open the cover plate.



Do not discard the PCS door panel bolts casually. After wiring is completed, proceed with reassembly

6.2 Check Port Information

6.2.1 Check DC Device Port Information

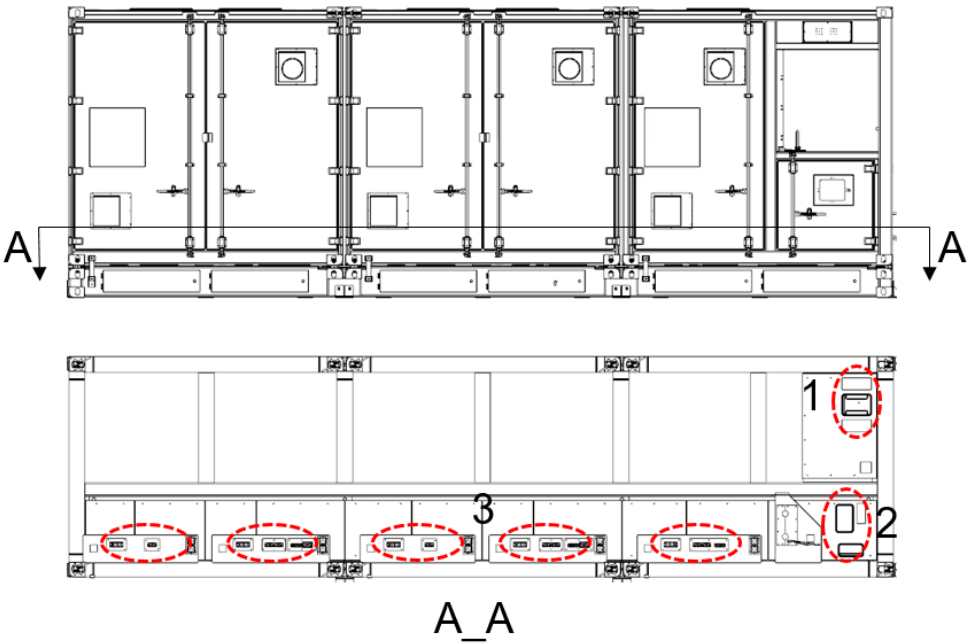
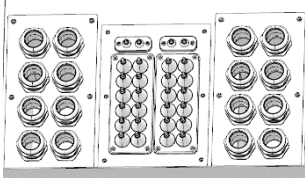
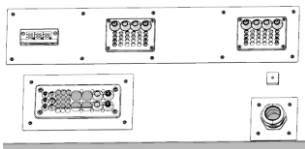
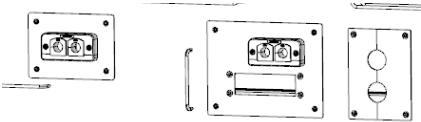


Figure 6-1 Wiring position of DC system

No.	Name	Cable	Picture
1	External power cable outlet of DC system	DC internal power cable, AC/DC power cable	
2	External control cable outlet of DC system	AC/DC control cable	

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No.	Name	Cable	Picture
3	Internal control cable outlet of DC system	RACK control cable and power cable	



The figures here are for your reference only. The actual products might differ based on the specific contracts.

6.2.2 Check AC Device Port Information

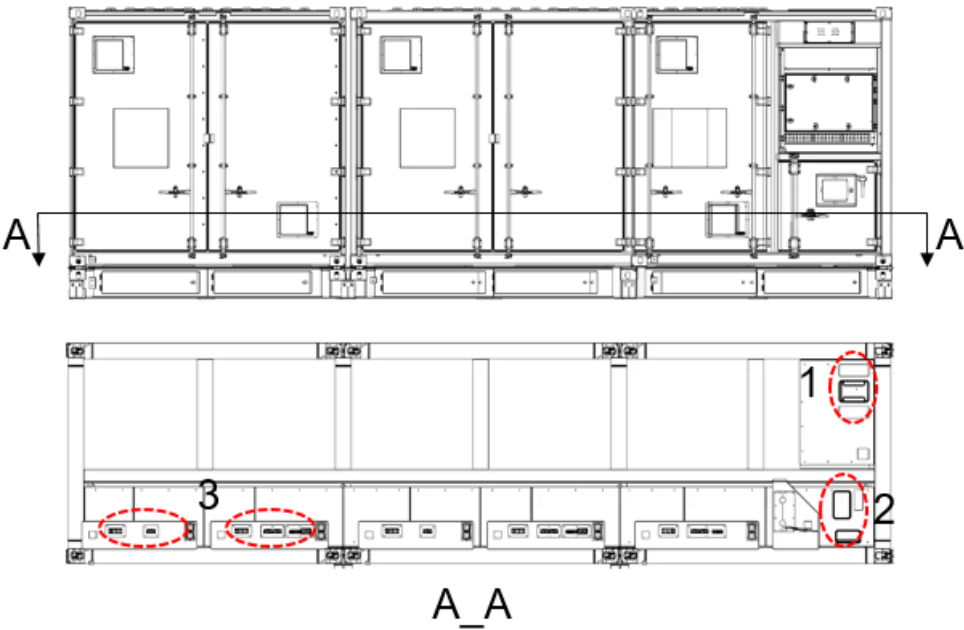
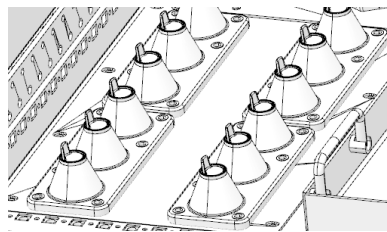
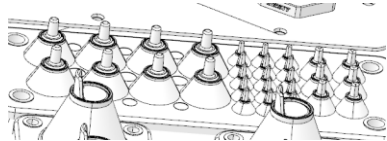
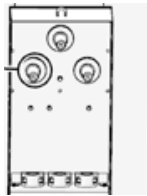


Figure 6-2 Wiring position of AC system

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S/N	Name	Cables	Figures
1	PCS power cable outlet	AC/DC system power cable	
2	PCS control cable outlet	AC/DC system control cable	
3	High-voltage cabinet outlet	33 kV cable	

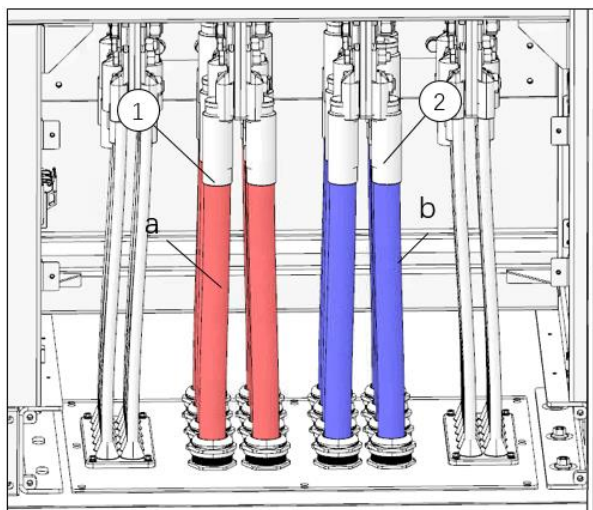


The figures here are for your reference only. The actual products may differ based on the specific contracts.

6.3 Disconnect DC-AC System Cables

6.3.1 Disconnect DC-AC System Power Cables

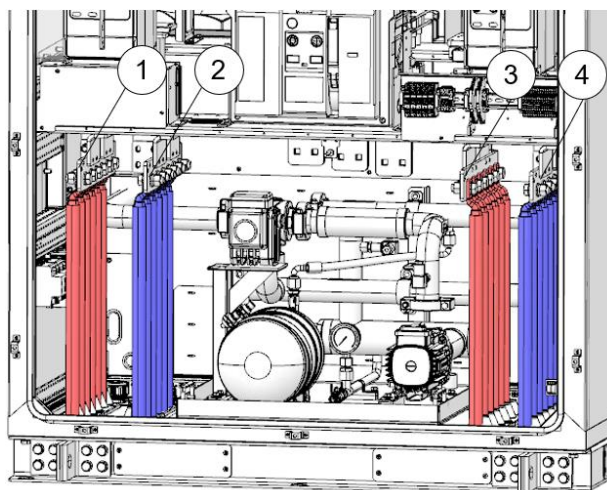
- 1 Remove the bolt assembly of the power cables from the distribution cabinet of the DC system, dismantling power cables.



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- ① PCS+
- ② PCS–
- a AC-DC power cable - positive
- b AC-DC power cable - negative

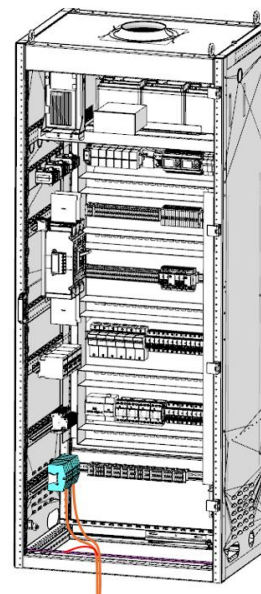
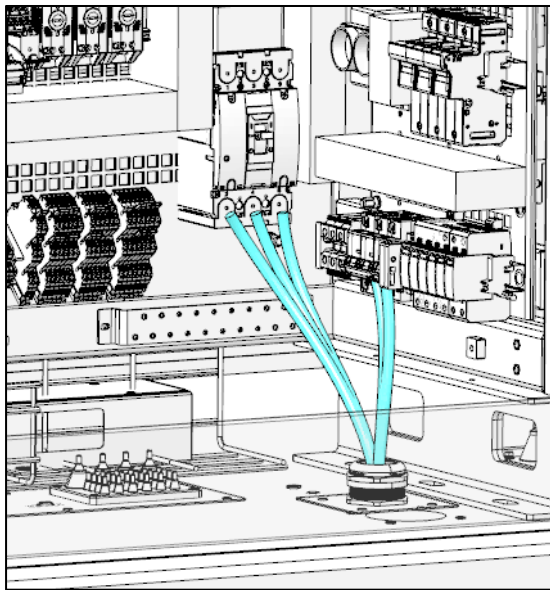
- 2 Remove the bolt assembly of the power cables from the distribution cabinet of the AC system, dismantling power cables.



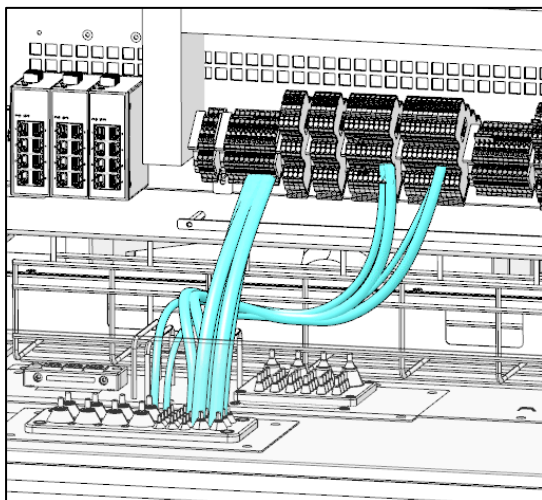
- ① DC1+
- ② DC1–
- ③ DC2+
- ④ DC2–

6.3.2 Disconnect DC-AC System Control Cables

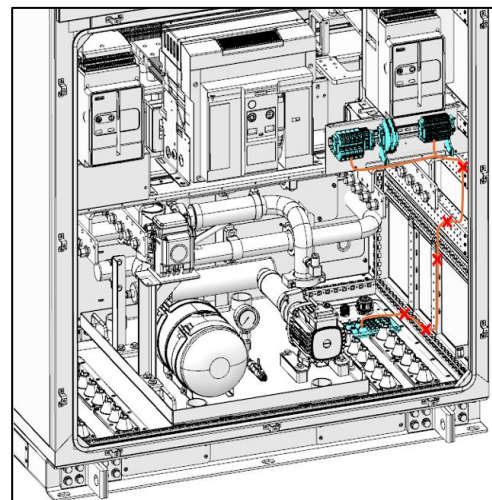
- 1 Remove the auxiliary power supply cables between AC and DC systems.



- 2 Remove the CAN/dry contact communication harness between AC and DC systems.

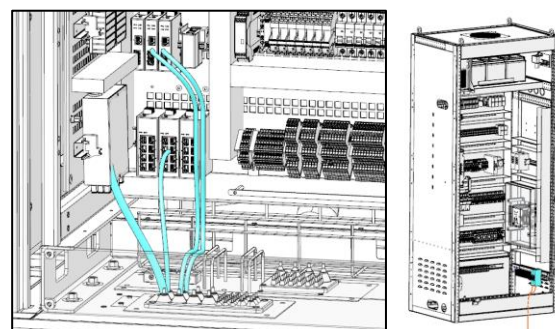


DC side



AC side

- 3 Remove the network cables between the AC and DC systems.



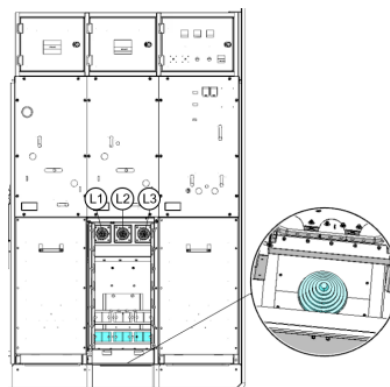
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6.3.3 Disconnect High-Voltage Cables

Remove the high-voltage cables of AC system.

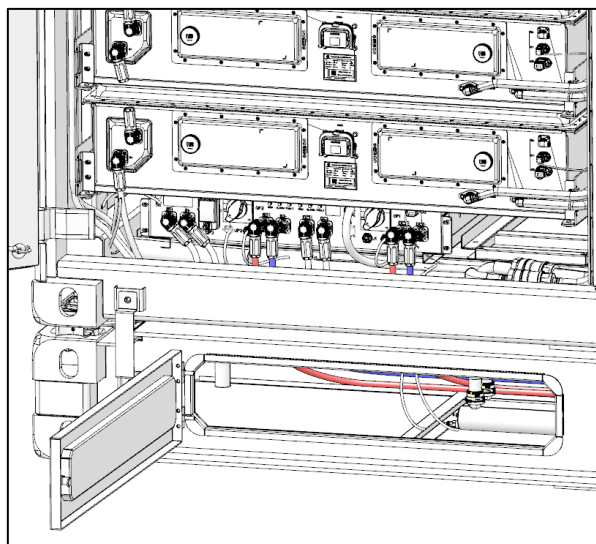


6.3.4 Disconnect BESS Station Cables

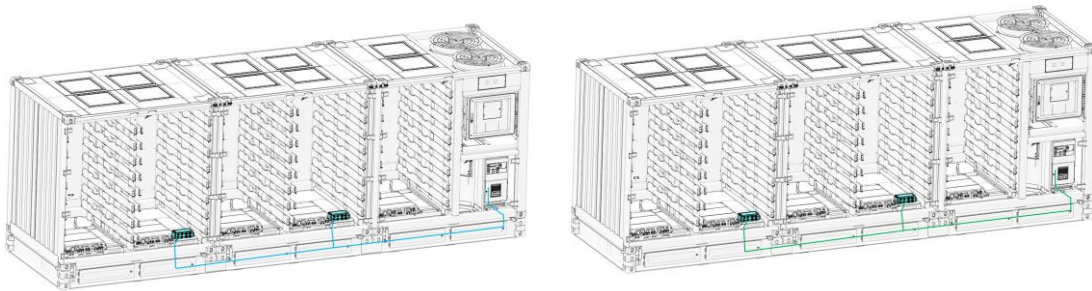
Remove the cables between the energy storage system and the station.

6.4 Disconnect DC system cables

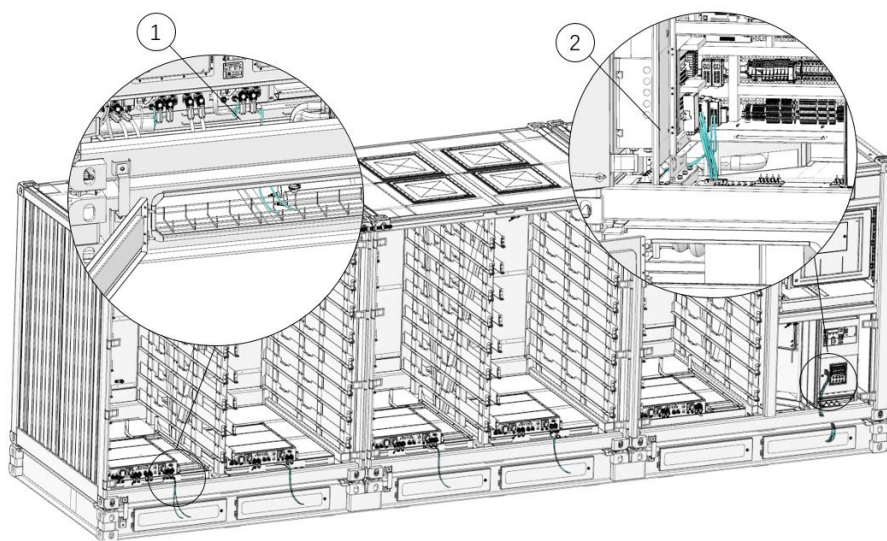
- 1 Remove the power cables inside the BESS system: Open the base cover plate under each RACK cabinet, and remove the cable connector of the high-voltage box above.



- 2 Remove the 24 V & communication harness for the junction box in container.



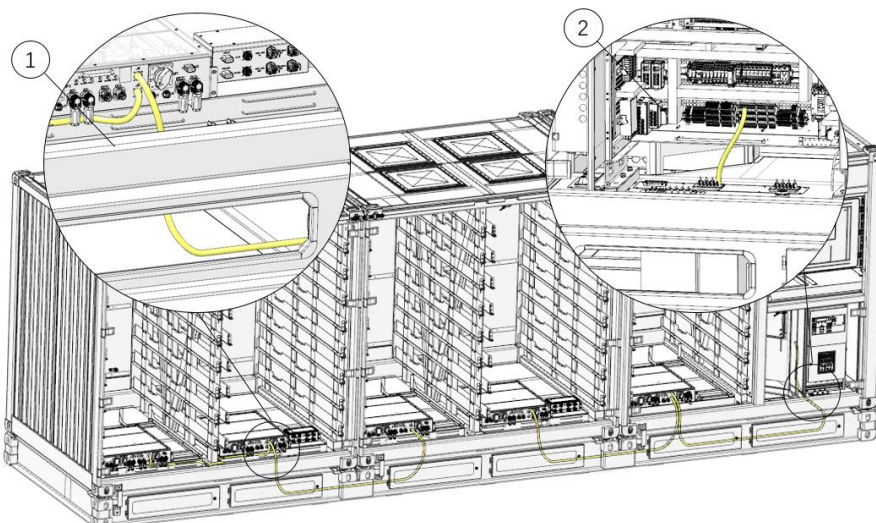
- 3 Remove the network cable from the RBBJ to the Power Distribution Cabinet.



① high-voltage box

② power distribution cabinet

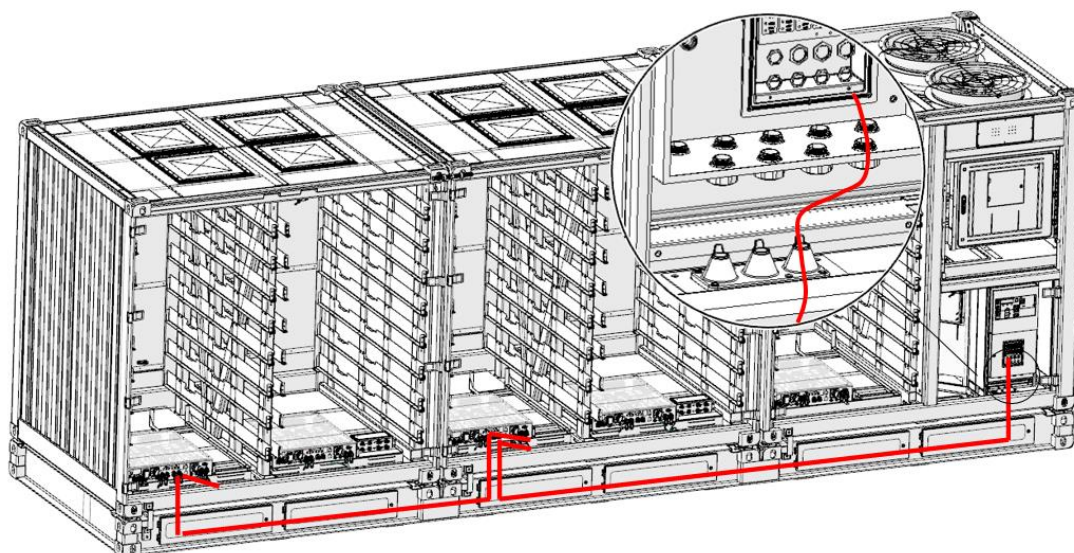
- 4 Remove the BMS harness from the RBBJ to the PDC.



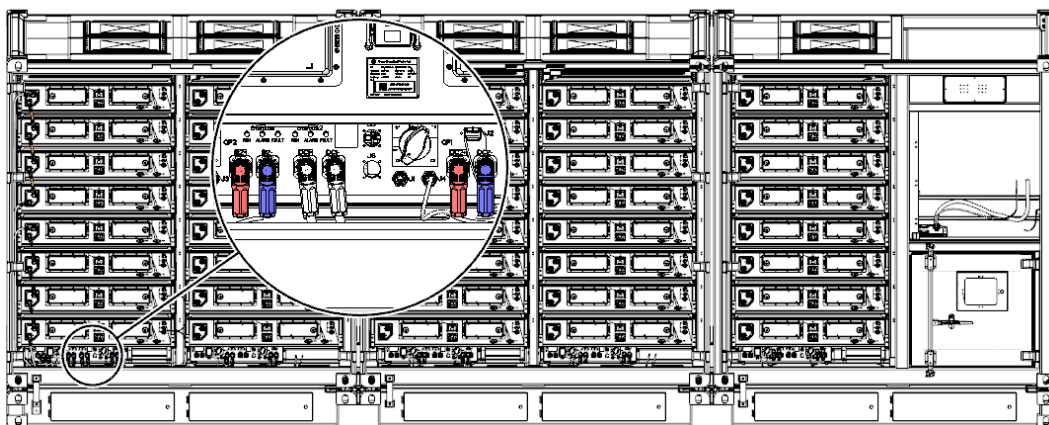
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- ① high-voltage box
- ② power distribution cabinet

5 Remove the fire protection cable in container.



6 Complete the dismantling of power cable connectors for each RACK within the DC system.

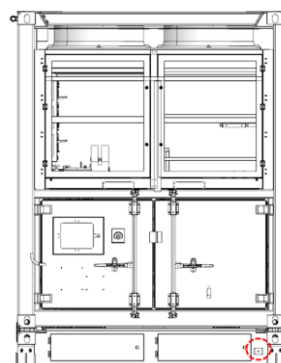
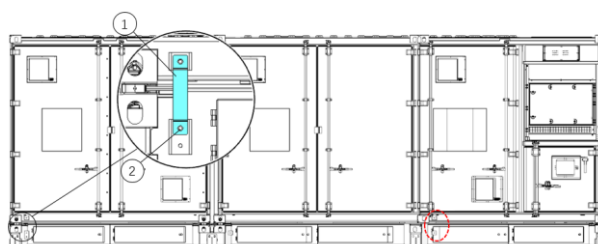


7 Remove The Grounding Connection Of The Equipment

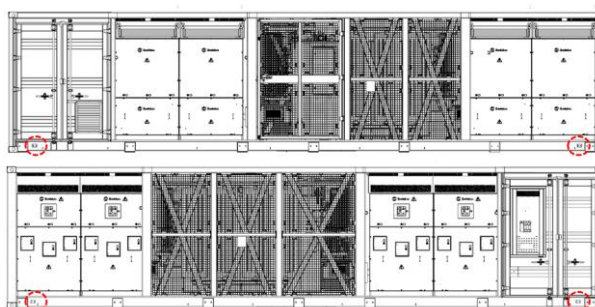
Before lifting the equipment, remove the grounding cable. One set of energy storage equipment has four grounding points on the DC side and four grounding points on the AC side.

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- 1 Remove the four grounding points on the DC system.



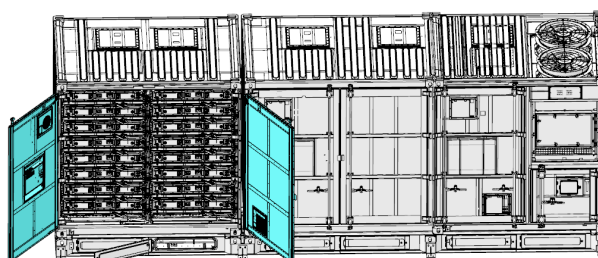
- 2 Remove the four grounding points on the AC system.



8 Remove Environmental Control Pipeline

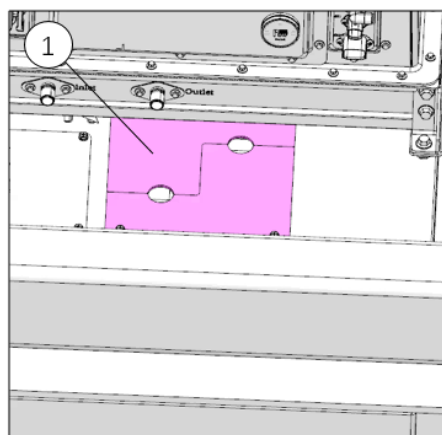
8.1 Remove RACK-Level Environmental Control Pipeline

- 1 Open each RACK door and the SKID cover plate.



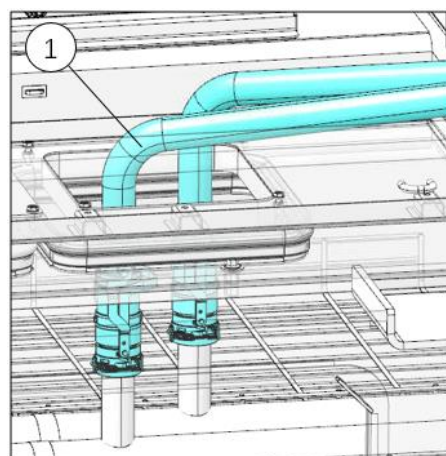
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- 2 Remove the bottom sealing plate of the RACK-level inlet and return water pipelines.



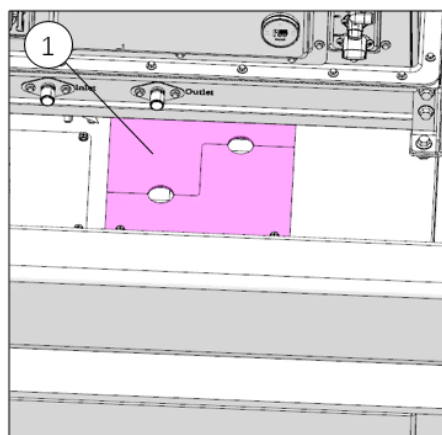
③ Bottom sealing plate

- 3 Remove the RACK-level inlet and return water pipelines as shown in the figure.



① water pipeline

- 4 Reinstall the bottom sealing plate.

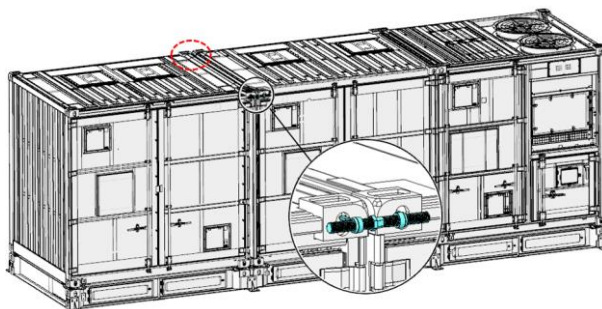


① Bottom sealing plate

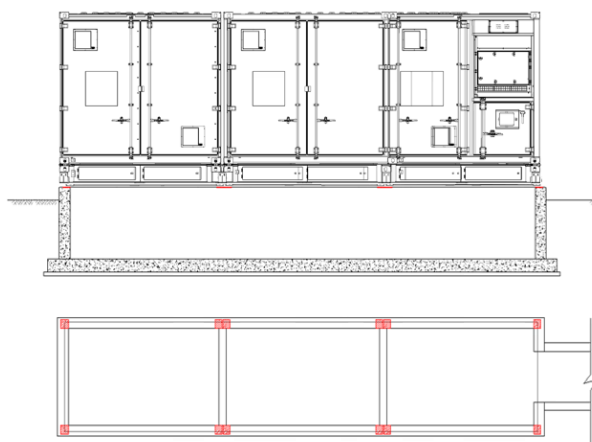
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9 Remove DC and AC System Anchoring Connection

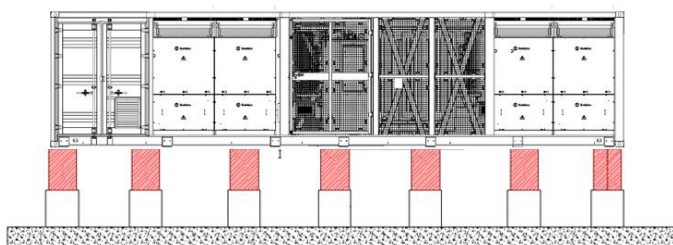
- 1 Use a step ladder to remove the fasteners on the container.



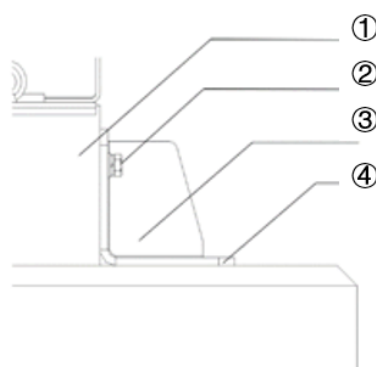
- 2 Remove the DC system anchoring connection as shown in the figure.



- 3 Remove the AC system anchoring connection as shown in the figure.



- 4 Equipment base connection type



① Equipment base

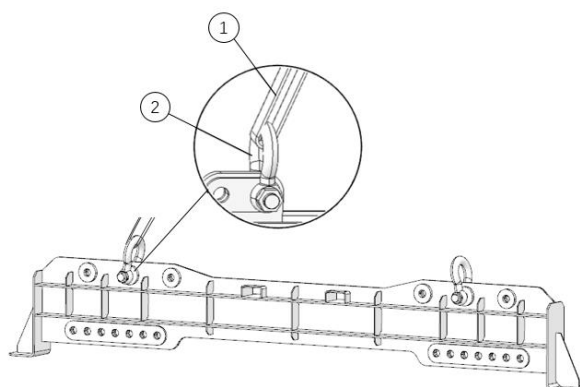
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- ② M16 bolt set
- ③ Angle iron
- ④ Connecting bolt

10 Lift DC and AC System

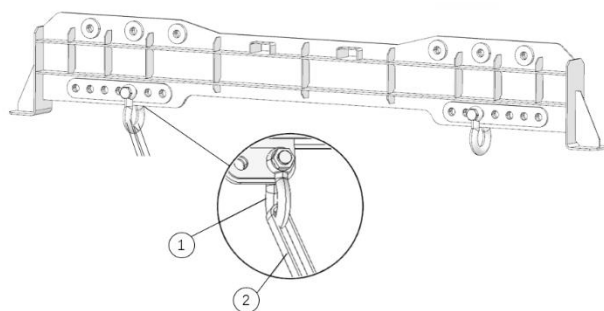
10.1 Install Lifting Tool Assembly

- 1 Use shackles, slings to install the upper slings of the lifting tool.



- ① Sling
- ② Shackle

- 2 Use shackles, slings to install the lower slings of the lifting tool.



- ① Shackle
- ② Sling

WARNING

- Wear protective gloves and safety shoes during tool assembly installation.
- Lifting equipment shall be kept stable during placement or assembly at all times to prevent it from tipping over and causing injury

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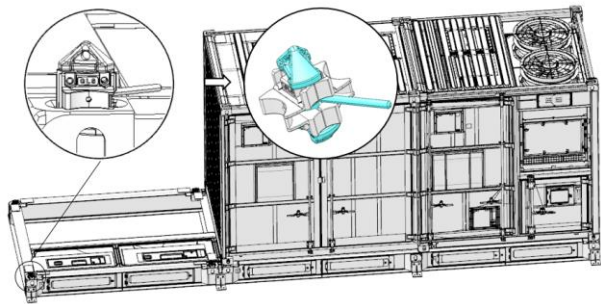


Note that the hanging point positions shown in the figure are for reference only. Refer to the actual hanging points of the product for installation.

- 3 Use different lifting points for different equipment.

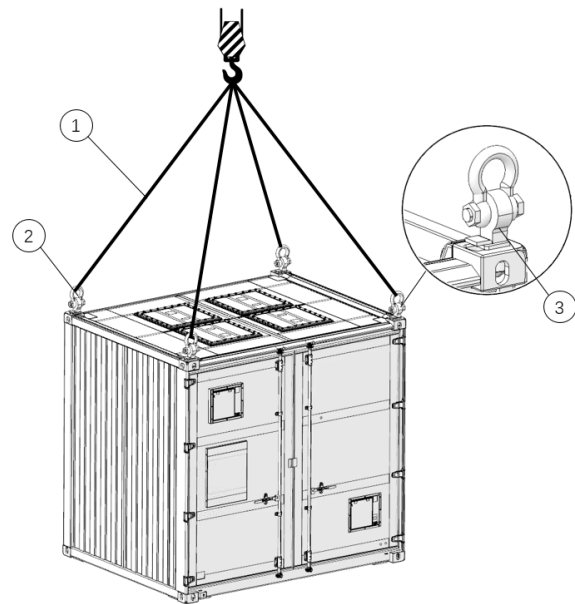
10.2 Lift DC System

- 1 Open the mushroom head fasteners between SKID and the container.



- 2 Hang the slings (for Battery Segment):
 - 1) Hang the four 15 t-9 m slings above the container.
 - 2) Connect the slings to the container with 17 t shackles and hooks.

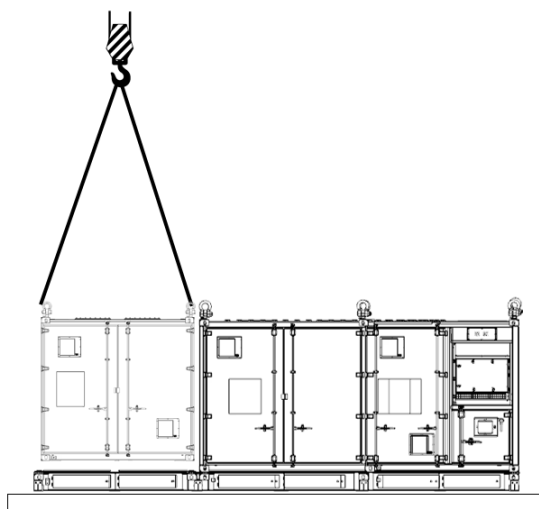
Tie a tagline to the lifting points on one side of the long edge of the container.



- ① 15 t-9 m sling
- ② 17 t shackle
- ③ Hook

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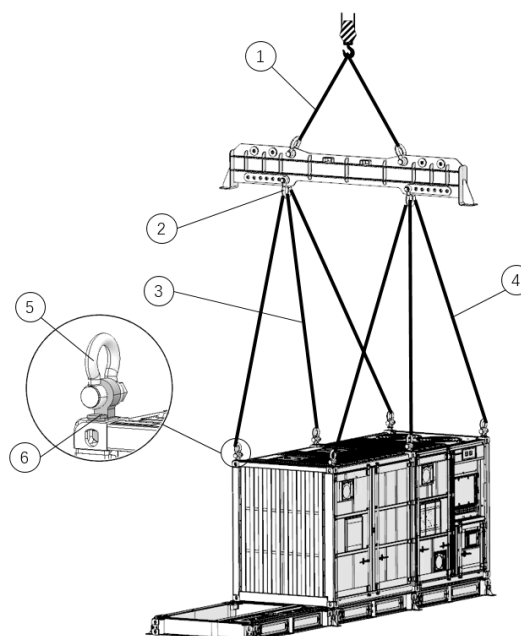
3 Lift Battery Segment from the SKID.



4 Hang the slings:

- 3) Hang the two 40 t-3 m slings above the container.
- 4) The tool and the upper and lower slings are connected by four 55t shackles.
- 5) Near the SKID end, use two 20t-15m slings (folded in half) to connect to the enclosure.
- 6) Near the main control cabinet end, use two 25t-8m slings to connect to the enclosure.
- 7) Connect the slings to the container with 17 t shackles and hooks.

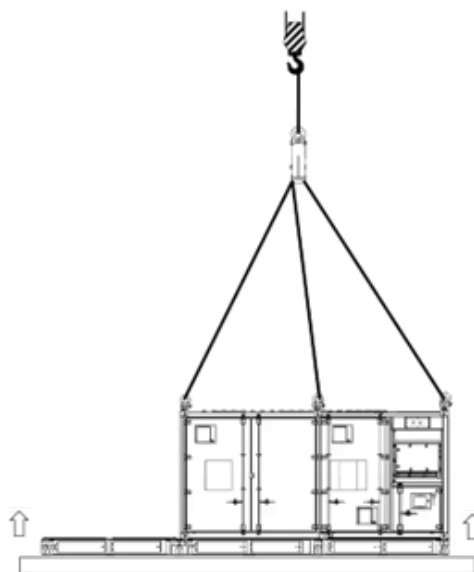
Tie a tagline to the lifting points on one side of the long edge of the container.



- ① 40 t-3 m sling
- ② 55 t shackle
- ③ 20t-15m sling(fold)
- ④ 25t-8m sling
- ⑤ 17t shackle
- ⑥ Container hook

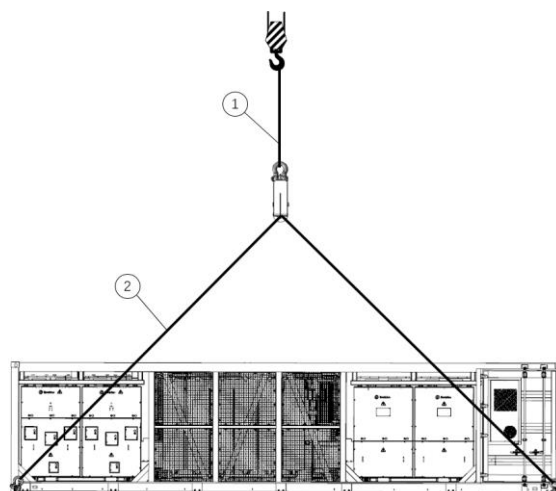
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- 5 Lift the container along with the SKID from the foundation



10.3 Lift AC System

- 1 Hang the slings:
 - 1) Hang the two 40t-3 m slings above the lifting tool.
 - 2) Hang the four 25t-9 m slings under the lifting tool.
 - 3) Connect the slings to the container body with 17 t shackles and hooks.
- Tie a tagline to the lifting points on one side of the long edge of the SKID.



① 40 t-3 m sling

② 25 t-9 m sling



Pay attention to adjusting the forward tilt through the hook and sling during the lifting process.

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Standing directly beneath the load is prohibited.

2 Lift the AC system from the SKID.

